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Recent Advancements in Wireline Microfrac Operation to Facilitate Successful Formation Breakdown – Experiences from Middle East and North Africa

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Abstract

This study presents the results and insights gained from recent wireline straddle packer microfrac testing conducted in tight Paleozoic and Mesozoic reservoirs, as well as caprocks in the Middle East and North Africa. A primary challenge encountered was the high breakdown pressure, which exceeded the tool's limits. To address this issue, the pump-out method was extensively used to achieve successful breakdowns in these tight formations, which exhibited an average formation breakdown pressure gradient of up to 1.16 PSI/ft, surpassing the vertical stress gradient. The pump-out method was implemented when the first injection cycle failed to achieve breakdown within the tool's pressure limit. We combined cyclic injection (hole pressurization) with pump-out (hole depressurization below mud hydrostatic) to successfully provoke breakdown. The back-to-back loading and unloading cycles created significant radial stress variations in the near-wellbore region, leading to fatigue and rock weakening. The pump-out technique works by destabilizing the mud cake, which allows fluid to infiltrate the formation and promotes pressure communication between the wellbore and any fractures that may have initiated. This study highlights the proven advantages of the pump-out method, which facilitated successful formation breakdown in impermeable and robust formations. The loading-unloading technique combined with pump-out has proven to be an effective approach for achieving successful formation breakdown without compromising the integrity of the tools used.

Introduction

The minimum horizontal principal stress (S_{hmin}) is a critical parameter in geomechanical modeling. Accurate estimation of S_{hmin} is essential for casing depth selection, drilling mud weight optimization, stimulation design, fluid injectivity, and caprock integrity assessments, applicable for oil and gas, geothermal, and CO₂ sequestration. Among various fracture diagnostic parameters, fracture closure pressure (FCP) is regarded as the most precise indicator for S_{hmin} . The FCP corresponds to the closest internal pressure that is just sufficient to open a fracture. S_{hmin} is usually estimated from a leak-off test (LOT), extended leak-off test (XLOT), minifrac, diagnostic fracture injection test (DFIT), hydraulic fracturing

(HF), hydraulic testing of pre-existing fractures (HTPF), or relief methods (Zoback, 2007). To enhance the consistency and reliability of stress estimates, Morawetz et al. (2020) proposed a quality ranking scheme for S_{hmin} measurements. This framework assigns ‘A-quality’ ratings to tests that yield repeatable FCP values across three or more consecutive cycles, ensuring robustness and reproducibility.

Such reliable S_{hmin} measurements can be achieved in the downhole wireline microfrac tests using modular sampling and pressurizing tools with a straddle packer, which have gained attention due to their superior quality, accuracy, and repeatability compared to the traditional minifrac tests (Franquet, 2020). Wireline dual packer microfracturing tests are performed in open hole where a 3-5 ft formation interval is isolated by wireline straddle packers. A small volume of drilling mud (~7-30 liters) is then injected at a controlled low flow rate (~0.1-2 liters/minute) (Franquet, 2020). The tests usually consist of three or more injection and shut-in cycles, each consisting of formation breakdown or fracture reopening, followed by fracture propagation, and fracture closure. When conducted properly, such tests provide ‘A-quality’ S_{hmin} estimates. Without successful formation breakdown, fracture propagation and closure cannot be achieved, and thus, an unsuccessful breakdown translates to a failed microfrac test. Therefore, achieving a successful breakdown is a key element of a microfrac test. In this study, we present actual field test examples from the Middle East and North Africa, where high breakdown pressures (FBP) were experienced, and breakdown could not be achieved in the first injection cycle of the microfrac tests. Based on the field data and lessons learnt from the recent tests, we discuss the remedial solutions and recent advancements that enabled the successful completion of the tests.

Challenges related to high breakdown pressure

Formation breakdown, unlike minimum horizontal stress (S_{hmin}), is not an inherent property of the subsurface formation but rather a response to various external influences that alter the near-wellbore stress field or the hydraulic connectivity between the wellbore and fracture systems. Several key operational parameters, including injection rate, wellbore diameter, and drilling fluid characteristics, significantly impact breakdown pressure, thereby influencing fracture initiation and formation breakdown (Carter, 1992; Detournay and Carbonell, 1997). Elevated breakdown pressures can result from drilling mud characteristics and interactions between drilling fluids and formation rock, which modify the near-wellbore pressure distribution and rock strength. Successful fracture breakdown is a prerequisite for determining S_{hmin} , as fracture closure cannot be measured without formation breakdown. While elevated breakdown pressures are advantageous for maintaining wellbore integrity during drilling, they pose challenges in hydraulic fracturing operations, particularly in low-permeability reservoirs, where significant injection pressures are required to exceed the formation’s breakdown threshold. In scenarios where breakdown pressure surpasses equipment/tool capacity, certain intervals may become inaccessible for completion, resulting in data acquisition limitations and potential production losses.

Tackling high breakdown pressure

High breakdown pressure remains a persistent challenge in microfrac operations as well as conventional hydraulic fracturing, particularly in low-permeability formations with high compressive strength. Khan et al. (2025) nicely summarized the influencing factors, reduction strategies and research gaps to tackle high breakdown pressure in hydraulic fracturing. This issue is prevalent in tight reservoirs where significant injection pressure is required to initiate and propagate fractures. Addressing this challenge requires optimizing tool selection, refining operational parameters, and integrating alternative stimulation techniques. One of the key factors influencing breakdown pressure is wellbore diameter, which directly governs the injection capacity of downhole tools. Research indicates that smaller hole diameters provide higher injection pressure limits, improving the likelihood of successful fracture initiation (Sen et al., 2024a-b). For instance, a 6-inch tool capacity surpasses that of an 8.5-inch tool, making smaller-diameter wells a

viable option for pressure optimization. However, implementing this approach requires modifications to the drilling plan, including the potential use of pilot holes for microfrac testing, which could increase operational costs if subsequent enlargements are necessary for completion. In larger wellbore configurations, alternative fracturing techniques have been proposed to overcome tool limitations. [Franquet et al. \(2024\)](#) suggested employing sleeve fracturing with a single packer system, as it enables higher injection pressures compared to conventional dual-packer tools. This method enhances fracture initiation efficiency while maintaining control over injection pressure thresholds. Additionally, [Franquet et al. \(2019\)](#) proposed another solution of increasing wellhead pressure above the drilling mud pressure after the packers are set downhole. This approach was successfully implemented in unconventional Haynesville shale formations, demonstrating its efficacy in mitigating breakdown pressure constraints. It is to be noted that raising mud weight or drilling a pilot hole for microfrac data collection may not always be feasible and practical due to cost implications and subsurface challenges, which necessitate improving tool capacity for a particular hole size as well as optimizing operational processes to tackle high breakdown pressure challenges. The other alternative methods involve the use of specialized fluids to reduce breakdown pressure. [Tariq et al. \(2021\)](#) discussed the use of thermochemical fluids to remove the filter cake, which has a positive effect in reducing breakdown pressure. In a recent work, [Khan et al. \(2024\)](#) reported the effectiveness of endothermic fluid that can reduce the breakdown pressure by 34%-53%. From conventional hydraulic fracturing experience, [Almarri et al. \(2022, 2023\)](#) presented the effectiveness of near-wellbore cooling in reducing breakdown pressure. It emphasizes the need for adaptive microfrac techniques tailored to subsurface conditions, particularly in tight formations where fracture initiation remains challenging. The integration of optimized tool selection, wellbore diameter considerations, and pressure enhancement methods is essential for improving the reliability of microfrac operations. These findings contribute to a broader understanding of geomechanical constraints and inform best practices for hydraulic stimulation in complex reservoirs.

Recent advancements: application of pump-out and cyclic loading-unloading

The cyclic injection method has been commonly discussed by many researchers as an effective method to weaken the rock and induce fatigue that facilitates breakdown ([Su et al. 2022](#); [Wu et al. 2023](#); [Mouli-Castillo et al. 2024](#)). In a wireline straddle packer microfrac test, this can be achieved by repeated injection cycles if breakdown is not achieved in the very first injection cycle. However, the maximum injection pressure would be guided by the maximum tool limit for that hole size. Based on the laboratory experiments, [Tariq et al. \(2021\)](#) suggested that mud cake removal lowers the breakdown pressure. In a microfrac operation, this could be achieved by the pump-out method, which helps remove some volume of mud filtrate from the near-wellbore region using a small-volume pump available in the microfrac tool string.

We combined the cyclic injection method and the pump-out method to achieve breakdown. The combination of rapid (using the maximum stepper of the injection tool) cyclic injection (hole pressurization), followed by pump-out (hole depressurization), results in loading and unloading of stresses that induce fatigue in the formation. The loading above the mud hydrostatic level during injection and unloading below the hydrostatic level during pump-out generate a significant radial stress variation, facilitating near-wellbore fatigue breakdown. For isotropic elastic homogeneous rocks, borehole stresses are given by the classical Kirsch elastic solution ([Kirsch 1898](#)) or its generalized version for nonaligned borehole and stress directions. [Fig. 1](#) illustrates the graphical representation of induced stresses and rapid changes in radial stresses during injection (in red) and pump-out (dark blue) while fatiguing the formation. The solution indicates an increase of radial stress σ_{rr} (compression) and a decrease of tangential stress $\sigma_{\theta\theta}$ when the wellbore pressure between the packer (P_w) increases during fluid injection for micro-fracturing. Conversely, radial stress (σ_{rr}) decreases (tension), and the tangential stress ($\sigma_{\theta\theta}$) increases during pump-out as P_w drops.

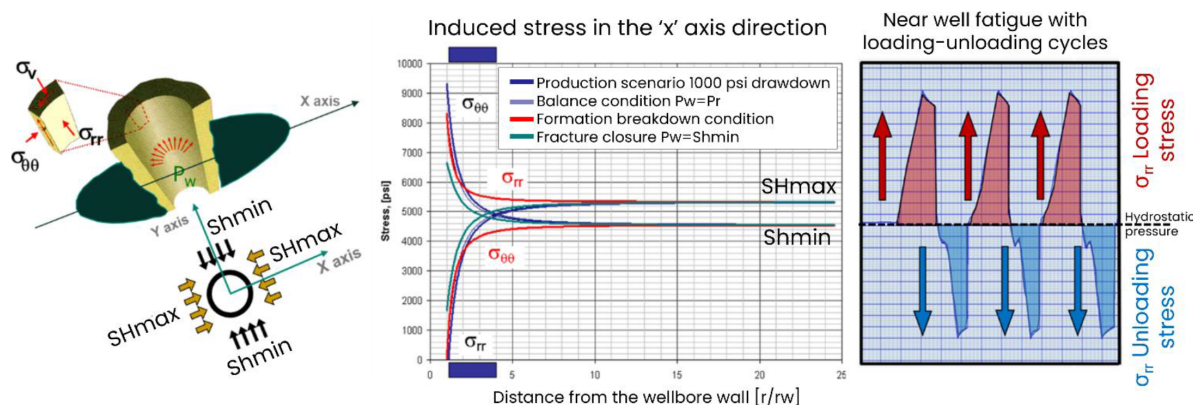


Figure 1—Graphical representation of the near-borehole stresses during cyclic loading and unloading approach for achieving borehole mechanical fatigue and formation breakdown.

The operational workflow of this approach starts with an interval selection, which would be isolated using dual packers over a 3-5 ft interval. In the injection phase formation is pressurized to the maximum tool capacity using max stepper of the pump. If breakdown is not achieved, we hold the maximum pressure for 2-3 minutes, and then a pump-out procedure is initiated to depressurize the near-wellbore region below the mud hydrostatic level. Usually, 5-15 liters fluid volume is pumped out during the unloading process, however, it greatly depends on the formation tightness. If pressure drops too quickly, then the unload cycle involves a limited or short pump-out cycle. This cyclic process of loading (pressurization) and unloading (pump-out) continues until the breakdown is achieved, as confirmed by changes in pressure. At extremely tight rock, a greater number of loading-unloading cycles could be required before formation breaks down. Real-time monitoring of the cycles and interpretation of the pressure behaviour are critical for operational decisions that facilitate the deployment and continuation of this approach. In the next section, we present the three field examples where breakdown could not be achieved in the first injection cycles and therefore loading-unloading approach was deployed to break the rock.

Field examples

Case-1: Breakdown after two loading-unloading cycles. Fig. 2 presents a microfracturing test performed in a tight Triassic quartz arenite interval with 6% porosity at a depth of ~3838m. The formation pore pressure was 8170 psi, and a downhole overbalance of 300 psi was maintained by OBM. The interval was quite strong with an estimated unconfined compressive strength (UCS) of 15500 psi. Breakdown could not be achieved in the first injection cycle, and therefore, the pump-out approach was deployed. During the first two injection cycles (cycle-1 and cycle-2), the wellbore was pressurized to the maximum tool limit (~14400 psi, which is 5900 psi above the mud hydrostatic level), followed by depressurization via pump-out (5500 psi below hydrostatic during pump-out-1 and pump-out-2). This process facilitated a breakdown at 1560 psi below the tool limit. The back-to-back loading-unloading cycles created a total radial stress change of 11400 psi in the near wellbore region during Cycle-1 and Cycle-2, which effectively weakened the rock. The FBP gradient was inferred to be 1.1 psi/ft, exceeding the vertical stress (S_v) gradient. The third cycle provided a conclusive fracture closure pressure (FCP) gradient of 0.88 psi/ft, using flowback method.

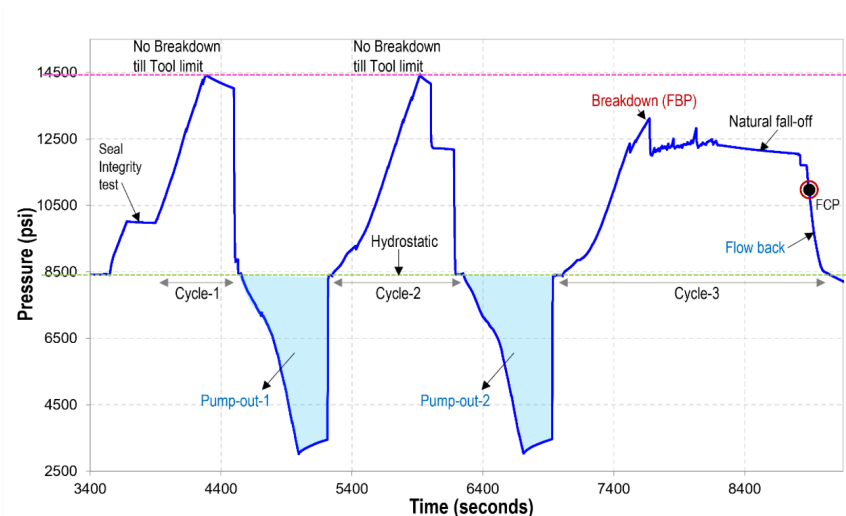


Figure 2—Microfrac test from a tight quartz arenite interval, showing two cycles of loading and unloading before successful breakdown in the third injection cycle. Green and pink horizontal lines indicate the mud hydrostatic pressure at the test interval TVD and tool limit, respectively.

Case-2: Breakdown after three loading-unloading cycles. Fig. 3 presents a microfrac data where three loading cycles and three pump-out cycles were applied. The test was performed in a Triassic quartzite reservoir interval at 3865m. The target formation had 7% porosity and ~0.08 mD permeability. The formation pore pressure was 8370 psi, and mud hydrostatic pressure was 8685 psi (315 psi downhole overbalance using OBM). The interval was very strong, with an estimated UCS of 15000 psi. At this test depth, breakdown could not be achieved in the first three injection cycles, even with downhole pressure reaching the tool limit. However, FBP was achieved 250 psi below the tool limit in the fourth injection cycle, after three rapid loading-unloading (injection and pump-out) cycles. The injection cycles (Cycle-1 to Cycle-3) pressurized the wellbore to 6000 psi above the mud hydrostatic, while the pump-outs (Pump-out-1 to Pump-out-3) depressurized the wellbore to 5300 psi below hydrostatic, resulting in a total radial stress change of 11300 psi in the near wellbore region. The FBP gradient in Cycle-4 was 1.12 psi/ft (> S_v). The fourth cycle provided a conclusive fracture closure pressure (FCP) gradient of 0.9 psi/ft, using the flowback method.

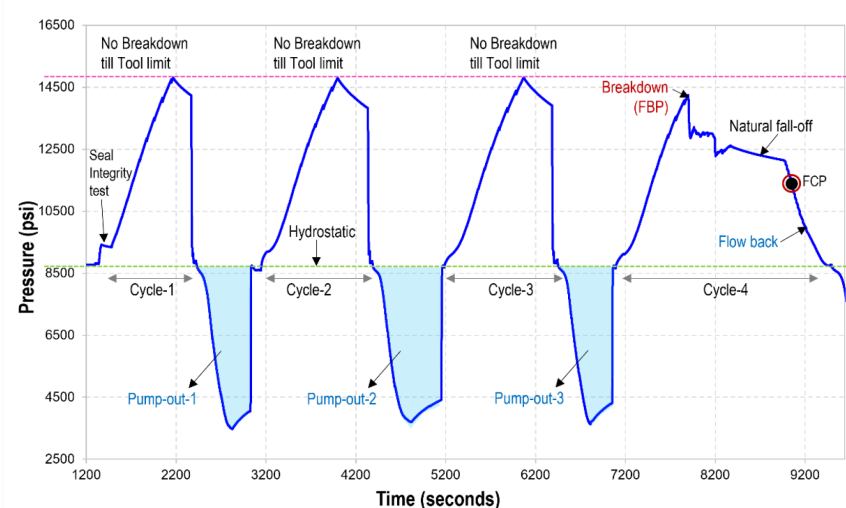


Figure 3—Microfrac test from a quartzite reservoir interval, showing three cycles of loading and unloading before successful breakdown.

Case-3: Extreme loading-unloading example. Fig. 4 presents an extreme example of a loading-unloading application for breakdown, performed in a tight crystalline dolomite interval at 3724m depth. The formation had 6% porosity and an estimated UCS of 14500 psi. A successful breakdown was achieved after five injection cycles (pressurization by 6100 psi above hydrostatic level) and four pump-out cycles (depressurization by 5100 psi below hydrostatic), which created a total radial stress change of 11200 psi in the near wellbore region. FBP was achieved in the sixth injection cycle, close to the tool limit. The inferred FBP gradient was 1.16 psi/ft ($> S_v$).

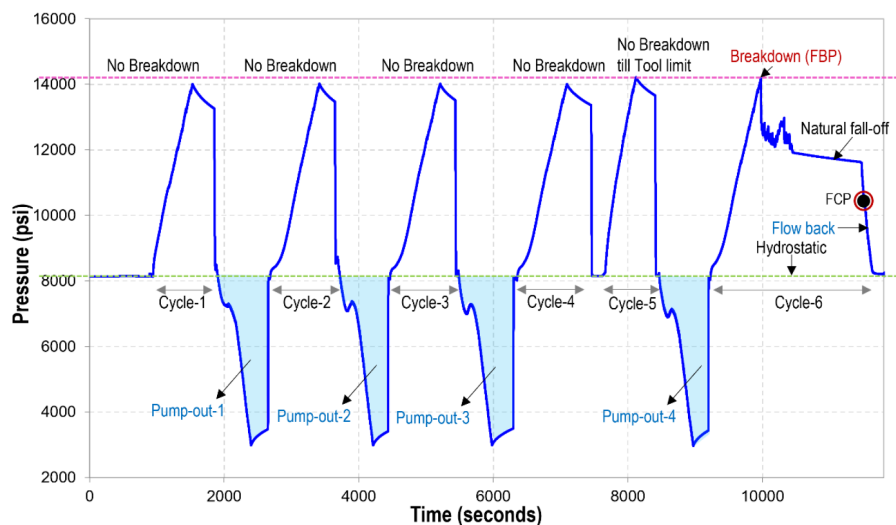


Figure 4—Microfrac test from a tight dolomite target, exhibiting an extreme case of five loading and four unloading cycles. A successful breakdown was achieved in the sixth injection cycle.

Discussions

We have summarized the formation properties and test parameters of the three field cases in Table 1. The field results elucidate the effectiveness of combining loading (injection) and unloading (pump-out) cycles in facilitating the breakdown of geological formations. The pump-out operation plays a critical role during shut-in phases by enhancing formation leak-off, thereby increasing the reproducibility of fracture closure measurements. While the field examples demonstrate successful breakdown and reliable closure pressure assessments through the techniques discussed, it is essential to recognize that multiple mechanisms may be influencing these outcomes. The rapid cyclic nature of loading and unloading may weaken the surrounding rock, inducing fatigue that could subsequently lower the breakdown pressure. Although fatigue and stress corrosion are established mechanisms, it is pertinent to note that these phenomena can occur independently of pressure cycling. Time-dependent fracture initiation and fatigue mechanisms are well-documented in the literature, as shown by Lu et al. (2020). Candela et al. (2014) showed that dynamic stressing at higher rates can enhance permeability by unclogging the pore system. The same mechanism could be a play in the loading-unloading cycles.

Table 1—Summary of the microfrac tests discussed, highlighting the key formation characteristics and operational parameters.

Case	Formation	Depth (m)	Porosity (%)	UCS (psi)	No. of loading cycles	No. of unloading cycles	Total radial stress change by loading-unloading (psi)	Injection cycle number when FBP achieved	FBP gradient (psi/ft)	FCP gradient (psi/ft)	Reference figure
1	Triassic Quatz Arenite	3838	6	15500	2	3	11400	Cycle-3	1.1	0.88	Fig. 2
2	Triassic Quartzite	3865	7	15000	3	3	11300	Cycle-4	1.12	0.9	Fig. 3
3	Crystalline Dolomite	3724	6	14500	5	4	11200	Cycle-6	1.16	0.9	Fig. 4

Additionally, the breakdown pressure can be influenced significantly by the characteristics of the drilling fluid employed. Power and Zamora (2003) posited that the formation of a mud cake, with its inherently low permeability, may isolate formation fluid from wellbore fluid, increase the pore pressure locally in the near-wellbore region, and prevent the development of tensile stresses in near-wellbore areas. Similarly, investigations by Feng et al. (2018) and Cook et al. (2016) highlighted the strengthening effect of mud cakes, which contributes to an increase in breakdown pressure. Gan (2013) observed that a higher solid content in drilling fluid yields a stiffer and more robust filter cake, resulting in elevated compressive stresses in the near-wellbore region. Under such conditions, the formation breakdown pressure is elevated due in part to the effects of the mud cake. Karpov et al. (2017) documented a pertinent field case from West Siberia indicating that the application of loss control materials (LCM) in drilling mud has been associated with increased formation breakdown pressure. Furthermore, mud invasion and the presence of mud filtrate can significantly alter the pressure within the formation, thereby inducing poroelastic effects and stress perturbations in the near-wellbore region. As a result, higher injection pressures may be required to achieve breakdown. The implications of mud cake effects are substantial, as they can fundamentally alter the dynamics within the near-wellbore environment, significantly impacting formation breakdown and fluid leak-off. This study emphasizes the potential variety of underlying mechanisms affecting breakdown pressure. The field test results suggest that the pump-out technique functions by destabilizing the mud cake, facilitating fluid infiltration into the formation, and/or enabling pressure communication between the wellbore and nascent fractures.

The presented microfracturing tests were conducted in the tight and competent formations. This study illustrates the substantial benefits of employing multiple cycles of loading and unloading through rapid injection (hole pressurization) and pump-out (hole depressurization). These tests support the hypothesis that an induced fatigue approach can effectively weaken the near-wellbore area, resulting in successful breakdown. Near-wellbore fatigue breakdown emerges as a viable strategy for managing the complexities associated with competent rock formations characterized by high breakdown pressures that may exceed the capabilities of downhole tools, which can be attributed to both the intrinsic properties of the rock and extrinsic factors that artificially elevate breakdown pressure. However, such strategies are only recommended for tight and competent formations, as weaker formations may be susceptible to collapse due to shear failure during pump-out operations.

Conclusions

This study summarizes our field experiences from the recently concluded microfracturing campaign in northern Africa and the Middle East and presents valuable insights on tackling high breakdown pressure during the tests. These tight clastic and carbonate rocks in a strong strike-slip tectonic regime with high formation breakdown pressures exhibited significant challenges to the successful conclusion of wireline microfrac operations. Proactively assessing these challenges during the testing phase is vital, as real-time monitoring and interpretation can significantly improve the validity and usefulness of the results obtained. To minimize risks, it is crucial to closely monitor fluid injection, ensuring that subsurface pressure remains within the safe limits of the downhole tools. Our field results indicate that integrating loading (injection) and unloading (pump-out) cycles proves effective in facilitating formation breakdown. However, it is important to recognize that lowering borehole pressure between the packers during pump-out cycles may lead to hole failure or sanding, particularly in weak or poorly consolidated rock. Acknowledging this limitation can help refine our field workflow. By placing an emphasis on understanding rock strength during the pre-job phase, we can strengthen our approach to effective data acquisition and improve outcomes in future microfracturing operations.

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